

Reduced-Complexity Partial MIMO Equalization for Real-Time GPU-based 10-Mode Transmission over 26 km GI-FMF

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Abstract We investigate partial MIMO equalization in real-time 10-mode transmission over 26 km of FMF and demonstrate that the technique successfully tracks channel dynamics over time, enabling flexible performance-complexity reduction trade-offs in MG-selective FMF transmission. ©2026 The Author(s)

Introduction

Space division multiplexing (SDM) remains a promising solution for high-capacity optical networks, with strong potential for long-haul applications [1]. Although current forecasts for SDM focus on transmission over uncoupled 2-core and 4-core fibers, coupled transmission over few-mode fibers (FMFs) continues to attract significant interest as a means to achieve higher spatial density [2].

Transmission over FMFs is subject to random mode coupling and modal dispersion, usually addressed by multiple-input multiple-output (MIMO) equalizers built from finite impulse response (FIR) filters [3]. The computational complexity of MIMO equalizers scales with the link length and the number of modes, imposing constraints on both mode count and transmission reach [4, 5]. Therefore, the implementation of coupled high-mode-count SDM systems is often considered impractical, even over relatively short distances.

Capacity records of SDM transmission have been reported based on the offline processing of large datasets, where the complexity of the digital signal processing (DSP) is not a primary constraint [6]. In contrast, reported real-time SDM demonstrations have pointed out challenges regarding speed, bandwidth, and precision that strongly limit the transmission baud rate [7–10].

To enable practical SDM transmission over FMFs, advanced fiber manufacturing processes and improved practices for device sharing and component integration must be complemented by novel reduced-complexity DSP techniques. Leveraging the limited coupling between modes of different mode groups (MGs) in graded index (GI) FMFs, a partial MIMO equalization approach has been proposed to reduce the number of FIR filters in MG selective transmission over 15 modes [11–13] and 55 modes [14]. To achieve a finer-grain data rate versus complexity trade-off, a rate-adaptive partial MIMO equalization scheme was studied in [15] for 15-mode transmission over 50 km of FMF.

So far, partial-MIMO equalization has been

demonstrated using offline processing of data traces acquired by digital sampling oscilloscopes, whose limited memory restricts the capture of the channel dynamics to a few microseconds. Since partial MIMO equalization relies on the time-varying channel coupling profile induced by environmental conditions, the evaluation of the technique beyond the memory capacity of oscilloscopes is key to assessing its practical feasibility.

This work demonstrates, for the first time, reduced-complexity partial MIMO equalization in a 10-mode GPU-based real-time transmission over 26 km of GI-FMF permanently exposed to environmental perturbations such as laboratory fans, rack vibrations, and foot traffic. Different MIMO input configurations are evaluated, and the performance is assessed over 120-second duration data traces. The results show that the MG selectivity of the GI-FMF is preserved over time, enabling different performance–complexity trade-offs through the strategic application of partial-MIMO equalization across the 4 MGs.

Experimental setup

The experimental setup of the real-time 10-mode transmission is depicted in Fig.1(a) [9]. An NKT Koheras BASIK fiber laser at 1550 nm was used for homodyne detection, eliminating frequency offsets and reducing the impact of laser phase noise on the low-baud rate signal. At the transmitter, an Xilinx RFSoc FPGA generated the two QPSK signals at 1 GBaud. Root-Raised-Cosine (RRC) pulse shaping was applied with 0.01 roll-off. The optical signal at the output of a dual-polarization IQ modulator was amplified and sent to a split and delay module to produce 10 decorrelated signals with a separation of 50 ns. The 10 optical signals were amplified and injected into a 15-mode multi-plane light conversion (MPLC)-based multiplexer [16]. A single span of 26 km of 15-mode GI-FMF supporting 9 linear polarized (LP) modes in 5 mode groups was used [17]. At the receiver, another identical 15-mode MPLC was utilized to

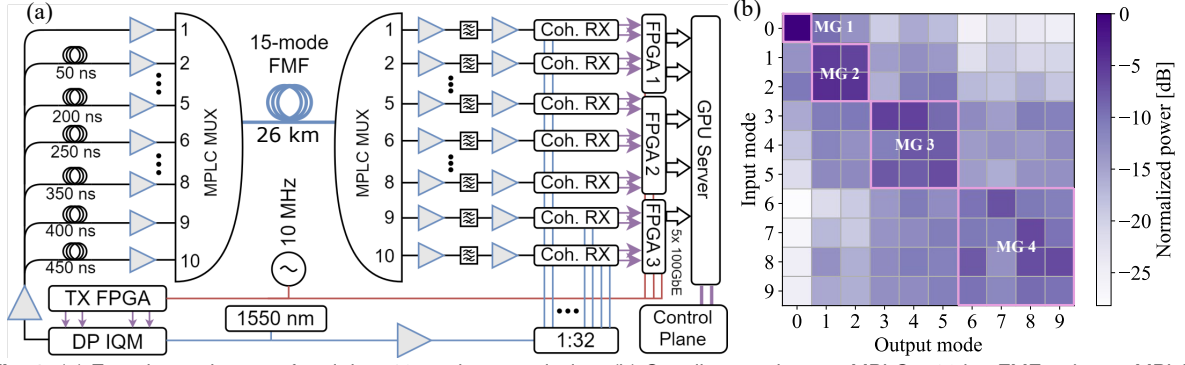


Fig. 1: (a) Experimental setup of real-time 10-mode transmission. (b) Coupling matrix: mux MPLC + 26 km FMF + demux MPLC.

demultiplex the spatial channels, and the first 10 modes corresponding to MGs 1 through 4 were amplified and forwarded to the receiver, where they were converted into the electrical domain using one 180° optical hybrid followed by balanced heterodyne detection per polarization channel. The receiver encompassed three Xilinx RFSoc FPGA ZCU216 evaluation boards, with each FPGA terminating up to 8 channels. All FPGAs shared a common 10 MHz reference clock, and a common trigger signal was used to synchronously start acquisition. Fine alignment was performed using individual programmable buffers placed into the FPGA. After alignment, the data was transferred using 5x 100Gbit/s Ethernet links to a server based on two AMD EPYC 7313 16-core processors and four NVIDIA A100 80 GB GPUs. All the details about the DSP chain performed by the transmitter and receiver FPGAs and the communication with the GPU server can be found in [9].

At the received DSP, a real-time block-based data-aided LMS complex-valued partial MIMO equalizer with 10 T/2-spaced taps and integrated phase recovery was used. The phase tracker operates on blocks of 32 symbols. Only two rows of the equalizer that recover both polarizations of 1 spatial mode at the output were implemented at once to ease resource requirements.

Experimental results

The power transfer matrix of the link that includes both MPLCs and 26 km of FMF is shown in Fig.1(b). The propagation of 10 spatial modes is done over 4 MGs defined as in the matrix. The power transfer between consecutive MGs goes

approximately from -3 dB to -10 dB, while decreasing to -25 dB between MGs at the extremes. The confinement of most of the energy into modes within the same MG indicates a mode group selective transmission where the signal power is concentrated around the main diagonal with decreasing coupling levels to more distant modes.

In an MG-selective transmission of $2N$ signals, where N denotes the number of spatial channels, reduced-complexity MIMO equalization can be achieved through the strategic deactivation of FIR filters, instead of employing a full $2N \times 2N$ MIMO equalizer with $(2N)^2$ FIRs filters [14]. Given the lower coupling levels between certain MGs, the transmitted modes can be recovered at the MIMO output using a reduced set of input modes, thereby decreasing the number of active FIR filters during equalization. Fig.2(a) shows the MG averaged SNR in dB as a function of the number of MGs used for equalization at the MIMO input to recover each one of the 4 MGs using real-time 0.002 millisecond data traces. For the 4 MGs transmission, each MG is recovered at the MIMO output using a single MG (target), 2 MGs, 3 MGs, and 4 MGs (full MIMO). In the case of using 2 MGs, there are two possible configurations to get MG2 and MG3, as indicated in the figure. The input MGs are selected by starting from the target MG and then extending toward neighboring MGs. For MGs 1 and 5, neighbors are selected in a single direction (either increasing or decreasing). For MGs 2 and 3, neighbors are selected in both directions. The curves in Fig.2(a) indicate a gradual increase in the SNR with the number of input MGs used for equalization, although MG2 and MG3 are more im-

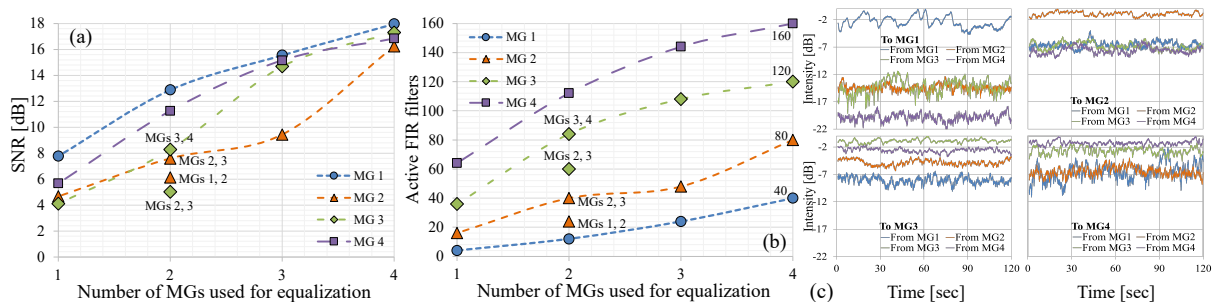


Fig. 2: (a) MG averaged SNR versus the number of MGs used at the input of the MIMO equalizer. (b) Active FIR filters versus the number of MGs used at the input of the MIMO equalizer. (c) Power transfer among MGs over 120 seconds from the equalizer taps.

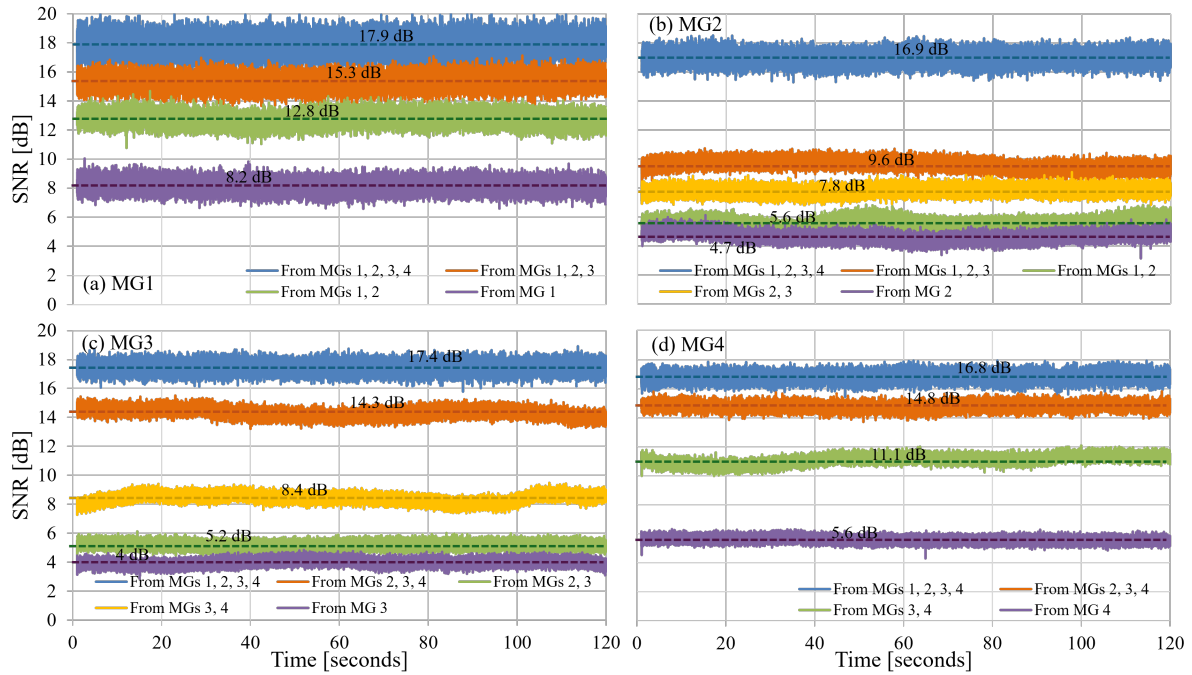


Fig. 3: MG averaged SNR in dB as a function of time in seconds while recovering (a) MG1 (b) MG2 (c) MG3 and (d) MG4 from different combinations of input MGs.

pacted by the removal of the MGs at the extremes due to stronger coupling levels. Recovering MG2 with 3 MGs (1, 2, 3) severely impairs performance since the energy transfer between MG2 and MG4 is non-negligible. Fig.2(b) presents the MIMO complexity in terms of active FIR filters as a function of the number of mode groups used for equalization. Reduced complexity is achieved by decreasing the number of input MGs for all output MGs. At two input MGs, for MG2 and MG3 using MGs 1, 2 and MGs 2, 3, respectively, a lower complexity can be achieved at the cost of SNR degradation as illustrated in Fig.2(a).

In transmission over FMFs, the channel response varies over time, making the coupling profile shown in Fig.1(b) also time-dependent. While partial MIMO equalization using 2-millisecond data traces provides valuable insight into the potential of the technique to reduce MIMO complexity, the performance must be monitored over longer periods to assess its practical feasibility.

To analyze the channel dynamics induced by laboratory fans, rack vibrations, and foot traffic, the power transfer among MGs, computed from the taps of the full MIMO equalizer, is monitored in real time over 120 seconds, as shown in Fig.2(c). The power leakage among MGs varies rapidly over time, following the expected coupling profile in MG-selective transmission over GI-FMFs, although some overlap is observed for some MGs.

Partial MIMO equalization results in real-time 10-mode transmission are shown in Fig.3 over 120 seconds. For all MGs, the SNR obtained from different sets of MGs remains stable over time, exhibiting minor short-term oscillations while stay-

ing close to the average indicated by the dashed horizontal line. In Fig.3(a), MG1 is the least affected by the removal of neighboring MGs, with SNR penalties of 2.6 dB, 5.1 dB, and 9.7 dB when using 3, 2, and 1 MGs out of 4 for equalization, respectively. However, MG1 allows for the lowest MIMO complexity reduction, as shown in Fig.2(b). In contrast, MG4 is the second least affected MG under partial equalization, offering the highest potential for complexity reduction. It should be noted that some energy from MG4 is coupled to MG5, which is not considered in this experiment. For MG2, the removal of MG4 is particularly detrimental, resulting in a 7.3 dB SNR degradation when going from full MIMO to using three input MGs. For MG3, using two input MGs remains conservative, while removing MGs 1 and 2 already incurs a 9 dB SNR penalty with respect to full MIMO. Notably, MG2 and MG3 experience less degradation when the lower-index neighboring MGs are removed rather than the higher-index ones. From Fig.3, partial MIMO-equalization is able to successfully track the channel dynamics and provide flexible performance-capacity trade-offs that are expected to be more efficient for stronger MG-selectivity.

Conclusions

This paper demonstrates partial-MIMO equalization in a real-time 10-mode transmission over 26 km of FMF subject to environmental perturbations. Performance monitoring over 120 seconds shows that partial equalization effectively tracks the channel dynamics while providing reduced MIMO complexity at the expense of a performance penalty in mode group-selective FMF transmission, enabling a new optimization degree in SDM systems.

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